Land Survey

Engineering Mathematics I for Land Survey

Trainer Guide - Keep Private

This guide gives marking expectations. Do not upload publicly if learners should not access marking guidance.

Marking Guidance

- Formula selection: 1–2 marks.
- Correct substitution: 1–2 marks.
- Accurate computation: 2–4 marks.
- Correct units: 1 mark.
- Technical interpretation: 1–2 marks.

Expected Response Pattern

1. For field distances, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in land survey context.

2. For coordinate differences, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in land survey context.

3. For bearings, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in land survey context.

4. For heighting, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in land survey context.

5. For control points, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in land survey context.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.